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ICS 435

ICS435 Final Project Report

GitHub Repository: https://github.com/erickimtypes/ICS435-Gene-Expression-Drug-Response-ML

1. Problem Introduction

The objective of this project is to predict how cancer cell lines respond to drug treatments using gene expression data. Specifically, this project will model the relationship between a cell line's gene expression profile and its measured response to a given drug.

A cancer cell line refers to a population of cancer cells that have been grown and maintained in a laboratory setting. These cell lines are commonly used in biomedical research because they provide a consistent and reproducible way to study how cancer cells behave and respond to treatments.

Gene expression refers to the level of activity of a gene, indicating how much of a particular gene is being used to produce RNA or proteins within a cell. In this project, gene expression values serve as features that describe the biological state of each cancer cell line, which can influence how the cell responds to a drug.

One major application of this work is in precision medicine, where treatments are tailored to individual patients based on their biological characteristics. By predicting how cancer cells respond to specific drugs using gene expression data, machine learning models can help identify the most effective treatments for a given patient, reducing the need for trial-and-error approaches.

This problem can be formulated as a supervised learning regression task. The input to the model is a high-dimensional vector of gene expression values for a cancer cell line, and the output is a drug response value derived from the PRISM dataset (log fold change, LFC). An LFC of 0 indicates no change, while a positive LFC indicates more gene expression and a negative LFC indicates less gene expression.

Machine learning is well-suited for this problem because gene expression data is high-dimensional, nonlinear, and noisy. Traditional statistical methods struggle to capture these complex relationships, especially when the number of features greatly exceeds the number of samples, as it does here. ML models are better equipped to learn patterns in such data.

The training data comes from publicly available pharmacogenomic datasets including gene expression data from the DepMap/CCLE project, and drug response data from the PRISM repurposing dataset.

Model performance was evaluated using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). RMSE measures the average magnitude of prediction error, with lower values indicating better performance. It is particularly useful in this context because it penalizes larger prediction errors more heavily. R^2 measures the proportion of variance in the drug response that is explained by the model, with values closer to 1 indicating better fit, and values near 0 indicating that the model performs similarly to predicting the mean.

In the context of biological data, such as gene expression and drug response, achieving very high R^2 values is typically difficult due to noise, measurement variability, and complex underlying biological processes. As a result, moderate R^2 values (e.g., around 0.3–0.6) can still indicate meaningful predictive performance. Similarly, RMSE values should be interpreted relative to the range and variability of the response variable, with lower values indicating more accurate predictions.

2. Data Introduction

The dataset used in this project is constructed by combining gene expression profiles with drug response measurements for cancer cell lines.

The gene expression data comes from the DepMap Cancer Cell Line Encyclopedia (CCLE), an open-access database from the Broad Institute that genetically and pharmacologically characterizes over 1,775 human cancer cell lines. In the dataset, titled

OmicsExpressionTPMLogp1HumanProteinCodingGenes.csv, each row represents a cancer cell (ModelID), and each column (total of 19,221 columns) represents a gene, with its values being log-transformed TPM expression levels. This results in an incredibly high-dimensional feature space that will need to be addressed. Additional metadata from *Model.csv* is used to identify and align cell lines across datasets.

The drug response data comes from the PRISM Repurposing dataset, a large-scale pharmacological screening resource developed by the Broad Institute. PRISM uses a pooled screening approach in which multiple cancer cell lines are treated simultaneously with barcoded identifiers, allowing high-throughput measurement of drug effects across hundreds of cell lines. This enables efficient profiling of drug sensitivity across a wide range of compounds and cancer types. In the dataset, titled *Repurposing_Public_24Q2_Extended_Primary_Data_Matrix.csv*, each of the 6,790 rows represents drug treatments, and each of the 920 columns represents cell lines (DepMap IDs). The values in the dataset are log fold change (LFC) measurements. Again, lower values of LFC correspond to stronger drug sensitivity, and higher values of LFC correspond to weaker responses or resistance.

To construct the final dataset, gene expression and drug response data are aligned using shared cell line identifiers. First, the overlapping set of cell lines is identified by the intersection of the ModelID column in the gene expression dataset, and the column names (cell lines) in the PRISM dataset. Only these 902 overlapping cell lines are retained for further analysis.

Next, the PRISM dataset is transformed from a wide format to a long format using a melt operation. This produced a table with columns IDs (drug identifier), ModelID (cell line), and Response (LFC value). This long-format dataset is then merged with drug metadata to obtain human-readable drug names.

For this project, the PRISM dataset is further restricted to a single drug, and rows with missing response values are removed. Finally, the filtered drug response data is merged with the gene expression data using an inner join on ModelID, producing the final dataset. In the final dataset, the features (X) are the gene expression values from the CCLE dataset, and the target (y) is the

LFC drug response value. Each sample corresponds to a cell line-drug pair, and all samples are labeled, since each includes a measured drug response value. However, the dataset remains relatively small compared to the number of features, resulting in a high-dimensional, low-sample-size setting, which is prone to overfitting

The drug response labels (LFC values) come from high-throughput experimental measurements and are widely used in pharmacogenomics research. While generally reliable, they may contain experimental noise, batch effects, and biological variability Thus, while the labels are trusted, they are not noise-free.

Both DepMap and PRISM are actively maintained datasets, and additional cell lines and drug response measurements are periodically released. Therefore, the dataset size is expected to increase in the future, which could improve model performance and generalization.

Key challenges in the data include high dimensionality, noise and biological variability, potential misalignment between datasets, and the risk of overfitting.

3. Model Introduction

To predict drug response from gene expression data, several supervised learning models were evaluated, ranging from linear to nonlinear approaches. Ridge regression was used as a baseline due to its L2 regularization, which helps mitigate overfitting in high-dimensional settings where the number of features exceeds the number of samples. To capture more complex, nonlinear relationships, Support Vector Regression (SVR), Random Forest, and XGBoost were also considered. These models were selected because they represent increasing levels of flexibility and capacity to model interactions within high-dimensional biological data.

All models were initially trained using baseline (default or minimally tuned) hyperparameters to establish a fair comparison. The dataset was split into 64% training, 16% validation, and 20% test sets to ensure that model selection and hyperparameter tuning were performed without using the test set, providing an unbiased estimate of generalization performance.

Model performance was evaluated on the validation set using Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). The baseline performance of each model is summarized in the table below. Random Forest, SVR, and XGBoost outperformed Ridge regression, indicating that nonlinear models better capture the relationship between gene expression and drug response. These models were therefore selected for further hyperparameter optimization, while Ridge regression was retained as a baseline for comparison.

Model	RMSE	R^2
Random Forest	0.970	0.354
SVR	0.940	0.392
XGBoost	1.04	0.260
Ridge	1.10	0.165

The results show the baseline performance of each model on the validation set. Once again, RMSE measures the average prediction error, with lower values indicating better performance, while R^2 measures how much of the variance in drug response is explained by the model, with higher values indicating better fit.

Among the models, Support Vector Regression achieved the lowest RMSE (0.940) and the highest R^2 (0.392), indicating the best overall predictive performance. Random Forest also performed relatively well, with an RMSE of 0.970 and R^2 of 0.354. In contrast, XGBoost and Ridge regression showed weaker performance, with higher RMSE values and lower R^2 scores, suggesting that they were less effective at capturing the relationship between gene expression and drug response under baseline settings.

Although the R^2 values may appear modest (R^2 ranges from 0 to 1, with higher values being considered better), this is expected in the context of biological data, where measurements are often noisy and influenced by complex, nonlinear processes. In this setting, R^2 values in the range of 0.2–0.4 can still indicate meaningful predictive signals. Overall, these results suggest

that nonlinear models, particularly SVR and Random Forest, are better suited for this task than linear approaches such as Ridge regression.

4. Hyperparameter Optimization

Hyperparameter tuning was conducted using Randomized Search with cross-validation (RandomizedSearchCV). This approach was chosen over grid search due to its efficiency in exploring a large hyperparameter space, particularly for computationally expensive models such as SVR and XGBoost. A fixed number of parameter combinations was sampled for each model, and performance was evaluated using 5-fold cross-validation (k = 5) on the training data.

For each model, the following hyperparameters were tuned:

- Random Forest: number of trees (n_estimators), maximum tree depth (max_depth), minimum samples required to split a node (min_samples_split), minimum samples required at a leaf node (min_samples_leaf), and the number of features considered at each split (max_features).
- Support Vector Regression (SVR): kernel type (kernel), regularization parameter (C), epsilon-insensitive loss parameter (epsilon), and kernel coefficient (gamma).
- XGBoost: number of boosting rounds (n_estimators), maximum tree depth (max_depth), learning rate (learning_rate), subsampling ratio (subsample), feature subsampling ratio (colsample_bytree), and regularization parameters (reg_alpha and reg_lambda).

During tuning, models were trained using the training set, and performance was evaluated using cross-validation within the training data. The primary scoring metric used for hyperparameter selection was negative mean squared error, corresponding to minimizing RMSE.

Model	Root Mean Squared Error (RMSE)
Tuned Random Forest	0.920
Tuned SVR	0.956
Tuned XGBoost	0.899

As shown in the results, the tuned XGBoost model achieved the best overall performance among all models, with the lowest RMSE. This indicates that XGBoost was most effective at capturing the complex, nonlinear relationships between gene expression and drug response.

The strong performance of XGBoost can be attributed to its optimized hyperparameter configuration. The best-performing model used a learning rate of 0.05, a maximum tree depth of 3, and 300 estimators, allowing the model to learn gradually while limiting model complexity to prevent overfitting. A subsampling ratio of 0.8 introduced randomness in the training data, improving generalization, while using all features at each split (colsample_bytree = 1.0) allowed the model to fully leverage the available gene expression information. Additionally, regularization parameters (reg_alpha = 0.1 and reg_lambda = 10) helped constrain the model, reducing the impact of noise and improving stability in the high-dimensional feature space. Together, these hyperparameter choices enabled XGBoost to effectively balance bias and variance, leading to improved predictive performance.

Overall, these results demonstrate that careful hyperparameter tuning can significantly improve model performance. In particular, the XGBoost model showed a substantial improvement, with RMSE decreasing from 1.04 in the baseline model to 0.899 after tuning. This reduction in error indicates that the tuned model is able to make more accurate predictions on average, capturing a greater portion of the underlying relationship between gene expression and drug response. In the context of noisy biological data, this level of improvement is meaningful, as even moderate reductions in prediction error can reflect a better ability to generalize beyond the training data. These results highlight the importance of hyperparameter optimization for flexible models such as XGBoost, which benefit from careful control of model complexity and regularization.

5. Evaluation and Results

The final tuned XGBoost model was evaluated on the held-out test set to assess its generalization performance. The model achieved a test RMSE of 0.825 and a test R^2 of 0.574. The RMSE of 0.825 indicates that, on average, the model's predictions differ from the true drug response values by approximately 0.825 LFC units. In other words, the typical prediction error is less than one unit of log fold change, suggesting reasonably accurate predictions given the variability of the data.

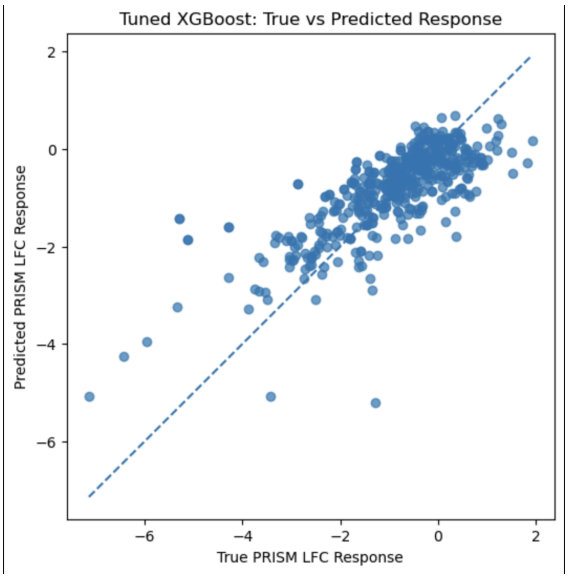
The R^2 value of 0.574 indicates that the model explains approximately 57.4% of the variance in drug response across the test set. This means that more than half of the variability in how cancer cell lines respond to the drug can be accounted for by their gene expression profiles as learned by the model.

Compared to the baseline XGBoost model, which achieved an RMSE of 1.04 and R^2 of 0.260 on the validation set, the tuned model demonstrates a significant reduction in prediction error and a large increase in explained variance. This improvement suggests that hyperparameter optimization enabled the model to better capture the underlying relationship between gene expression and drug response, while also improving its ability to generalize to unseen data.

The improved performance of the tuned XGBoost model, particularly on the held-out test set, indicates that the model is able to generalize beyond the training data. The use of cross-validation during hyperparameter tuning and the strict separation of training, validation, and test sets helped prevent overfitting and ensured an unbiased evaluation of performance.

However, several limitations remain. The dataset is relatively small compared to the number of features, increasing the risk of overfitting despite regularization. Additionally, gene expression and drug response data are inherently noisy and may contain measurement variability. There may also be distribution differences between cell line data and real patient tumors, which could affect the model's applicability in clinical settings.

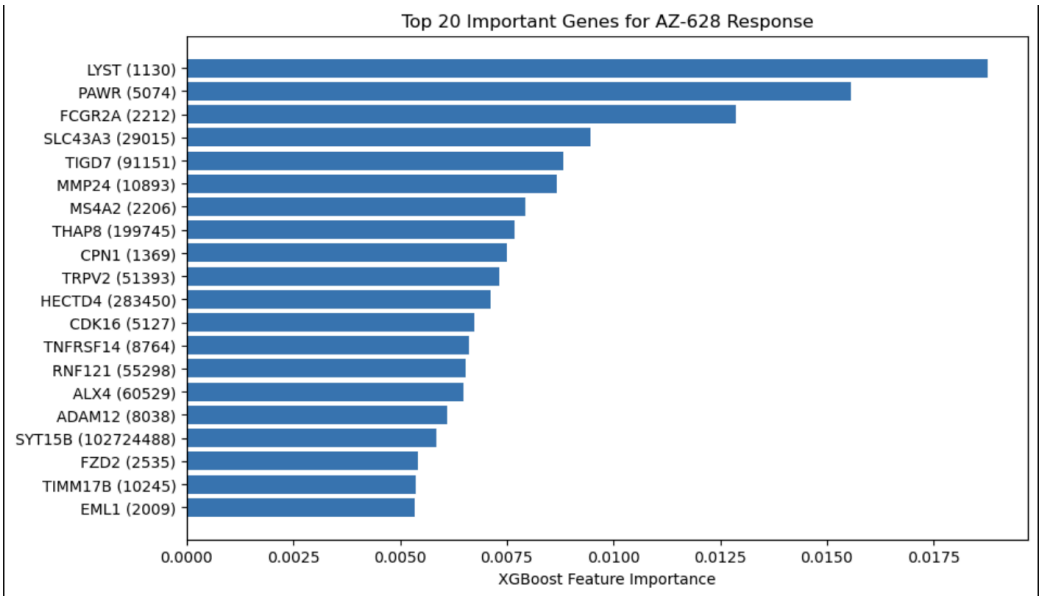
Despite these limitations, the results demonstrate that machine learning models, particularly well-tuned nonlinear models such as XGBoost, can capture meaningful patterns in gene expression data and provide useful predictions of drug response.



The relationship between predicted and true drug response values is visualized in this figure. In this scatter plot, each point represents a sample, with the x-axis corresponding to the true PRISM LFC response and the y-axis corresponding to the model's predicted response. The dashed diagonal line represents perfect predictions.

The plot shows a clear positive correlation between predicted and true values, indicating that the model is capturing the overall trend in the data. Many points lie close to the diagonal line, suggesting accurate predictions for a large portion of the dataset. However, there is still noticeable spread around the line, particularly for more extreme values, indicating that the model has greater difficulty accurately predicting outliers or highly sensitive/resistant responses.

This pattern is expected in high-dimensional biological data, where extreme responses are often more difficult to model due to noise and limited sample representation. Overall, the plot supports the quantitative results, demonstrating that the model generalizes well while still leaving room for improvement.



To better understand the model's behavior, feature importance scores from the tuned XGBoost model were analyzed. The figure above shows the top 20 most important genes contributing to the prediction of response to the drug AZ-628.

The results indicate that a relatively small subset of genes contributes most strongly to the model's predictions, despite the high dimensionality of the dataset. This suggests that the model is effectively identifying key features associated with drug response rather than relying on all genes equally.

Notably, genes such as LYST, PAWR, and FCGR2A were among the most influential features. While this analysis does not establish causal relationships, it highlights potential biological signals that may be associated with sensitivity or resistance to AZ-628. These findings could serve as a starting point for further biological investigation.

Several directions can be explored to extend this work. First, the model could be expanded to predict responses for multiple drugs simultaneously, rather than focusing on a single compound, allowing for broader applicability in drug screening. Incorporating additional data types, such as mutation, copy number variation, or proteomic data, could also improve predictive performance by providing a more complete representation of cellular biology. Furthermore, dimensionality

reduction or feature selection techniques could be explored to better handle the high-dimensional gene expression space and reduce noise. On the modeling side, more advanced approaches such as neural networks or ensemble methods combining multiple models could be investigated. Finally, validating the model on external datasets or patient-derived data would be an important step toward assessing its real-world applicability and potential use in precision medicine.